Kelly McManus

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Data engineer who owns the data platform and models that inform company decisions. From design to on-call, developed the pipelines behind Etsy's experimentation platform, supporting 700+ users and 2,500 experiments per year.

Work Experience

Etsy, Remote from Montréal, QC · *Marketplace for handmade and vintage goods*

Senior Software Engineer, Machine Learning I — Oct 2024 – Aug 2026

- Built the self-service platform that enabled experimenters to author 500+ metrics and 70+ segmentations via a SQL query by turning a merged pull request into a production Airflow DAG. Architected the Buildkite workflow that validated SQL definitions and automatically created 50+ parallel Airflow DAGs.
- Led a team of 3 scientists and engineers to scale variance reduction from 0.1% to 90% of metrics, enabling 10% more experiments to run each year. Designed the batch pipelines that prepared features and distributed linear regression with Spark. Drove the migration across ETL, Cloud Functions, frontend, and MySQL.
- Increased experiment results delivery SLA hit rate from 82% to 98%. Traced delays to a legacy Spark job. Reduced its runtime by 60% by retiring unused events instead of refactoring brittle Scala code.
- Saved $370K in annual GCP costs by building the first holistic view of experimentation compute and storage costs in Looker. Led 20+ incident responses and 2 company-wide post-mortems.

Software Engineer II — Feb 2023 – Oct 2024

- Created the Experiment Results Archive, extracting results from Firestore into structured BigQuery tables. Now the core tables behind Etsy's internal MCP server for experiment analysis, used by 150 monthly users.
- Implemented ML-based variance reduction in ETL to shorten experiments by 3 days, on average. Trained models with Vertex AI and built the Apache Beam inference pipeline. Published in Etsy's engineering blog.
- Built a Kafka streaming pipeline to sink experiment alerts to BigQuery, delivering real-time notifications to experimenters on Slack. Scoped requirements through 11 user interviews and an 8-squad pilot.

Data Engineering Intern, Experimentation — May 2022 – Aug 2022

- Extended the experimentation platform's statistical methods to three business areas running experiments off-platform: CRM, seller, and off-site ads. Unified experimentation methods across custom pipelines.

Bus.com, Montréal, QC · *Startup, marketplace for charter bus rentals* — Jul 2019 – Aug 2021

Data Analyst

- Owned warehouse connectors and analytics stack end-to-end: Fivetran ingested 12+ sources into BigQuery. Found PostgreSQL hard deletes did not propagate and updated the data model to remove deleted records.
- Built the LookML data model layer (explores, persistent derived tables), transforming BigQuery data for analysis in Looker. Trained marketing, finance, and product to perform self-service analysis.

District 3 Innovation Hub, Montréal, QC · *Nonprofit, startup incubator* — Jan 2017 – Jun 2019

Operations Coordinator, then Manager – Led the operations team behind validation and coaching programs, supporting 100+ startups. Systematized the application process and redesigned the database.

Education

Carnegie Mellon University, Pittsburgh, PA, USA — Sep 2021 – Dec 2022

Master of Information Systems Management, Business Intelligence and Data Analytics
GPA 4.0 / 4.3, Cofounder of Girls Who Code – Carnegie Mellon Chapter

Concordia University, Montréal, QC — Sep 2014 – Jun 2018

Bachelor of Commerce, Business Technology Management, Minor Data Intelligence

Skills

Languages: Python, SQL, Scala, Java. **Storage:** BigQuery, Iceberg, PostgreSQL, GCS. **Compute:** Spark, Apache Beam, Dataflow, Kafka. **Pipelines & Infra**: Airflow, Fivetran, Kubernetes, Terraform. **Observability**: Looker, Grafana.